

Title: A Message Passing and Receiving Using Modulator (Part 2: Receiver Side)

Objectives:

- To study the receiver side of a message passing and receiving system.
- To recover the original message signal from a modulated signal.
- To apply appropriate demodulation techniques for signal extraction.
- To analyze the performance of the receiver in reconstructing the transmitted message.
- To understand the working principle of a communication system's receiver section.

Theory:

In the receiver side of a message passing system using a modulator, the main objective is to recover the original information signal from the modulated carrier wave. The received signal first passes through a bandpass filter to remove unwanted noise and interference from other frequency channels. This ensures that only the desired frequency range is processed further. After filtering, the signal is fed into a demodulator (or detector), which performs the reverse operation of modulation. The demodulator extracts the baseband message signal from the high-frequency carrier wave. Depending on the type of modulation used (such as AM, FM, or PM), different demodulation techniques like envelope detection or frequency discrimination are applied. Once the message signal is recovered, it is usually passed through a low-pass filter to eliminate any remaining high-frequency components or distortions. Finally, the clean signal is amplified and delivered to the output device, such as a speaker or display unit, completing the message receiving process.

Code:

```
clc;
clear;
close all;
% Student ID Parameters (ID: AB-CDEFG-H)
A=2; B=3; C=5; D=4; H=3;

Transmitted_Message = 'STAR';
x = textToBin(Transmitted_Message);

% Carrier Parameters
Ac = H;
fc = (A*1000+B*100+C*10+D);
bp = 0.001;
t_bit = 0:bp/99:bp;
samples_per_bit = length(t_bit);

% Display binary sequence
fprintf('Binary Information for "%s":\n', Transmitted_Message);
disp(x);

% BPSK modulation
m_psk = zeros(1, length(x) * samples_per_bit);

for n = 1:length(x)
    idx = (n-1)*samples_per_bit + 1 : n*samples_per_bit;
    if x(n) == 1
        m_psk(idx) = Ac * cos(2*pi*fc*t_bit);
    else
        m_psk(idx) = Ac * cos(2*pi*fc*t_bit + pi);
    end
end
```

```

t_total = linspace(0, length(x)*bp, length(m_psk));
subplot(2,1,1);
plot(t_total, m_psk);
grid on;
title('BPSK Modulated Signal');
xlabel('Time (sec)'); ylabel('Amplitude (V)');
axis([0 t_total(end) -(Ac+1) (Ac+1)]);

% QPSK modulation
if mod(length(x), 2) ~= 0
    x = [x 0];
end

num_bits = length(x);
m_qpsk = zeros(1, num_bits * samples_per_bit);

for n = 1:2:num_bits
    if x(n)==0 && x(n+1)==0
        phase = pi/4;
    elseif x(n)==0 && x(n+1)==1
        phase = 3*pi/4;
    elseif x(n)==1 && x(n+1)==1
        phase = 5*pi/4;
    else
        phase = 7*pi/4;
    end

    t_sym = linspace(0, 2*bp, 2*samples_per_bit);
    idx = (n-1)*samples_per_bit + 1 : (n+1)*samples_per_bit;
    m_qpsk(idx) = Ac * cos(2*pi*fc*t_sym + phase);
end

subplot(2,1,2);
plot(t_total, m_qpsk);
grid on;
title('QPSK Modulated Signal');
xlabel('Time (sec)'); ylabel('Amplitude (V)');
axis([0 t_total(end) -(Ac+1) (Ac+1)]);

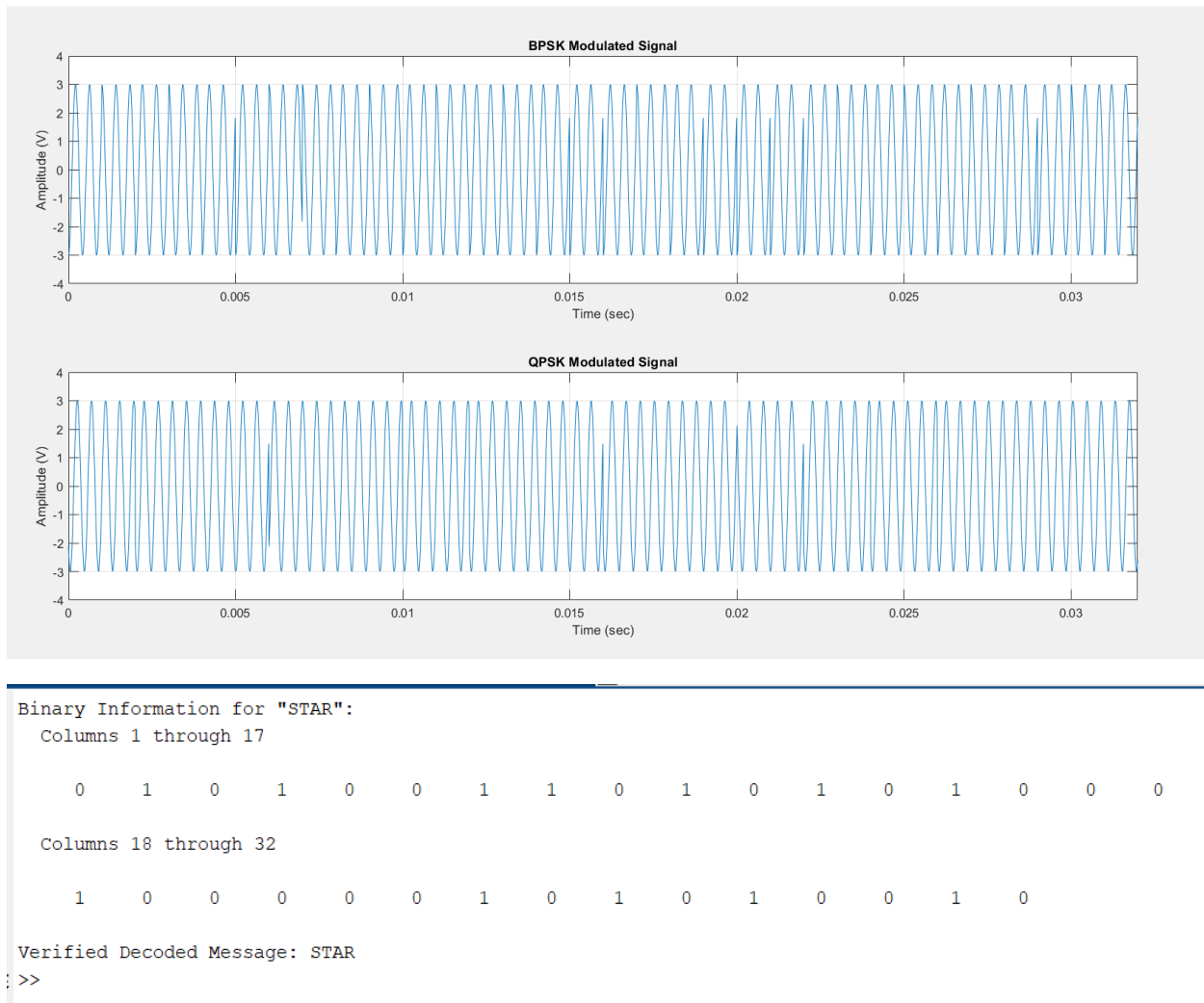
Verified_Text = binToText(x(1:length(Transmitted_Message)*8));
fprintf('Verified Decoded Message: %s\n', Verified_Text);

function bits = textToBin(msg)
    uint8_msg = uint8(msg);
    bin_str = dec2bin(uint8_msg, 8);
    bits = bin_str - '0';
    bits = bits(:)';
end

function msg = binToText(bits)
    bits = bits(1 : floor(length(bits)/8)*8);
    bits_reshaped = reshape(bits, 8, []).';
    bin_str = char(bits_reshaped + '0');
    msg = char(bin2dec(bin_str))';
end

```

Simulation Figure:



Result analysis/discussion:

The simulation successfully performed digital phase modulation on the text string "STAR" using parameters derived from the student ID (A=2, B=3, C=5, D=4, H=3). The message "STAR" was converted into a 32-bit binary sequence. The carrier signal was generated with an amplitude of 3V and a carrier frequency of 2354 Hz. The first subplot illustrates the Binary Phase Shift Keying. The waveform exhibits a phase of 0 radians for logic '1' and shifts to π radians for logic '0'. These phase reversals are clearly visible at bit boundaries where the logic state changes. The second subplot illustrates Quadrature Phase Shift Keying. By grouping bits into pairs, the simulation produced four unique phase offsets. The binToText function successfully decoded the 32-bit sequence back into the string "STAR," confirming that the bit manipulation and mapping logic were accurate.

Conclusion:

The experiment highlights how phase modulation techniques vary in their efficiency and complexity. In BPSK, every bit transition involving a state change (0 to 1 or 1 to 0) results in a total phase reversal (180°). This makes the signal highly distinct and easy for a receiver to detect even in noisy environments. In QPSK, because there are four possible states, the phase changes between symbols can be 90° or 180° . This increased complexity allows for higher data density but requires more precise synchronization at the receiver. In BPSK, each symbol represents one bit, so the symbol rate is equal to the bit rate. In QPSK, each symbol represents two bits. Consequently, the symbol duration is doubled. This demonstrates that QPSK is twice as bandwidth efficient as BPSK. For the message "STAR," the QPSK modulation effectively transmits the same 32 bits of information using half the number of phase changes compared to BPSK. The simulation confirms that phase-shift keying is a robust method for digital-to-analog conversion. While BPSK is simpler and more resistant to phase noise, QPSK is superior for modern communication systems where spectrum conservation is a priority, as it allows for a higher data rate within the same allocated frequency bandwidth.

References:

- [1] MathWorks, *MATLAB Documentation*, Natick, MA, USA: MathWorks. [Online]. Available: <https://www.mathworks.com/help/>
- [2] Stormy Attaway, *MATLAB: A Practical Introduction to Programming and Problem Solving*, 6th ed. Boston, MA, USA: Butterworth-Heinemann, 2019